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Autonomous Optimization at Scale: Leveraging AI for Closed-Loop, Real-Time Production Optimization

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Abstract

Gas lift is one of the most important methods of artificial lift for oil wells across the globe. However, the productivity of gas lift wells hinges on the fluctuating availability of compressed gas, presenting a complex optimization challenge. Traditionally, this issue has been tackled either manually, with significant delays, or automatically, with limiting assumptions that hinder optimal results. We introduce a groundbreaking closed-loop software solution that revolutionizes gas lift allocation by combining speed and precision.

This innovative software generates gas lift performance curves for each well using a hybrid approach incorporating both data-driven methods and physics-informed machine learning. These performance curves, along with critical physical constraints like minimum stable rates and maximum injection rates, enable the determination of optimal gas allocation strategies across varying gas availability scenarios. We also present a method for producing performance surfaces, which estimate production by considering choke position and flowline pressure in addition to gas lift injection rate, and thus allow optimization based on both gas availability and "take-away" constraints.

The implemented solution continuously estimates the available gas from the compressor system, based on real-time discharge pressure readings, and selects the most suitable strategy every 15 minutes for over 600 wells. Additionally, it provides insightful data on optimization success, offering a clear understanding of the system's performance. Furthermore, the highly responsive nature of the process has been shown to inherently maximize the utilization of the compression capacity installed. Remarkably, this approach has achieved up to a 4% production uplift in gas lifted wells, even before the implementation of the multi-dimensional production surfaces, which we expect to provide significant incremental value.

Introduction

From west Texas to the Middle East, gas lift wells make up an increasing share of oil production. Gas lift comes with many benefits: it is cost-effective to install and operate, robust to changing well conditions, and long-lived compared with other lift methods, particularly electrical submersible pumps (ESPs). Gas lift wells are so reliable, in fact, that they are often pushed to the back of the queue when troubles arise, with the understanding that they will soldier on while other systems might lose functionality without immediate attention; in some cases, wells can go months or even years without review or optimization (Gambaretto & Rashid, 2023). This is especially problematic in unconventional wells, where a well can have a substantial amount of decline over such a period (Ayoub et al, 2018), but also applies to conventional wells, especially as produced fluid fractions change.

With that problem in mind, various surveillance and optimization systems have been created to identify when a well is not receiving the optimal injection rate, and in some cases correct the issue (Ayoub et al; Borden et al; Masud et al). These systems can have a wide range of control capabilities, from surveillance-only to fully autonomous, with some intermediate systems providing recommendations that a human must approve. Another feature of these systems is the method in which information is generated: many rely on physical modeling, others on machine learning, and a few on analyses of direct measurements of production, or on a proxy such as down-hole pressure. One example is the system presented by ExxonMobil in 2024, which is a completely autonomous system relying primarily on automated analysis of direct measurements and falling back on machine learning when those measurements are suspect or unavailable (Movahed, 2024). Regardless of the optimization system applied to a given well, however, one limitation remains: to inject the optimal amount of gas into a well, there must be enough high- pressure gas available.

Superficially, the availability of gas for injection appears to be a non-issue: gas injected into a well is not consumed, but rather re-produced, so no loss of gas occurs in the process. All that is required is to install compression capacity equivalent to the injection requirement, and one might expect the question of gas availability to be solved. However, there are several factors that impact gas availability.

Throughout the industry's history, and especially in areas with high well counts, the required gas compression has been under installed in an attempt to avoid excessive capital expenditure. Furthermore, gas lift requirements tend to increase as wells age, turning what might once have been a lean design into a constrained one. Additionally, as an asset ages, the number of wells on gas lift increases, adding even more demand to the system than original designers might have planned for. Finally, and perhaps most importantly, gas compression is provided by mechanical systems which come with reliability concerns of their own: they cannot be run indefinitely and sometimes shut down without warning. This intermittency periodically reduces capacity but is often not considered substantial enough to warrant installing spare compression capacity because installing new compression is both expensive and operationally complex. For these and other reasons it frequently becomes necessary to operate wells, and thus also to optimize them, within the constraint of the gas available to the system rather than simply considering each well as an individual entity as previous optimization systems have.

System optimization is complicated by the transient nature of compression reliability issues. Additionally, there is significant variability in the duration of transient events. While both planned maintenance and unplanned interruptions tend to last multiple days, it is common for other unplanned compression interruptions to last only a few hours which is long enough to cause injection to start failing due to lack of pressure. In the case of short duration events, which are too short for a person to reoptimize the system manually, the system is allowed to operate sub-optimally while the compression events occur. To operate optimally during both long and short duration transient periods, a system is required that can rapidly identify changes in gas availability constraints and automatically optimize gas allocation as the constraint changes. We present a system doing exactly that at over 600 gas lift wells.

Methods

The presented software iteratively moves the system in the correct direction, focusing on rapidly responding in the optimal direction, rather than on achieving the exact optimal solution at each time step. We find that this focus on agility enables rapid response to highly variable events; the software can iterate as quickly as every 12 minutes, if needed, and that often the physical system has changed enough after 12 minutes that spending more time to identify a more precise solution would have been wasted. The method by which the software determines each steps, described in this section, considers each set of interconnected compressors and the wells they provide gas for as an independent system. This method has been easily scaled to dozens of such systems in the Permian Basin.

Machine Learning Gas Lift Performance Curves

For a system-level optimization to be done, a gas lift performance curve (GLPC) must be available. Most wells optimized by this software have GLPCs constructed based on results from automated multi-rate testing. This process cannot be applied at every well due to a mix of data issues, most commonly because the well does not have a functional down-hole pressure gauge. To enable optimization of systems where some wells do not have GLPCs, a set of physics-informed machine learning processes are leveraged to estimate GLPCs for the problematic wells.

Gas lift performance curves can be described reasonably well by a quadratic function, which requires three independent pieces of information to define. Identifying these begins with the calculation of a set of physics-based engineered features, such as the total gas-liquid ratio. These are fed, along with other metadata regarding the well, into a machine learning tool which estimates the optimal gas lift injection rate. This data is used to calculate, as described in detail in our previous paper (Movahed, 2024). This enables a relatively simple model (in this case, a random-forest model) to consider complex physical quantities without the need for extensive training data to learn the importance of the physical features from scratch. Since the GLPC is taken to be roughly quadratic, this injection rate defines the vertex, the slope of which is zero by definition. This point is the first of the three pieces of information required. Next, the injection rate is fed into a second model, in this case a LightGBM model, which estimates production rates considering gas injection and the information available for the well, such as previous well test rates, the time since that test was performed, and the reservoir zone (Nallapusala, 2025). This defines the production rate at the optimal injection rate, generating the second piece of information. Finally, a minimum stable rate is drawn from a mixture of data-driven analyses, physics-based modeling and operator feedback. The minimum stable rate is defined as the injection rate where the GLPC hits zero production. This, therefore, provides a third distinct piece of information.

These three pieces of information are then combined to calculate a quadratic approximation of a gas lift performance curve similar to that defined by multi-rate testing. This solution massively increases the robustness of the system-level optimizer, which would otherwise fail any time a gauge failed, and would remain unusable until the gauge is replaced, which typically does not occur until the gas lift system is pulled and replaced.

Gas Lift Availability Estimation

Any constraint-based optimization method naturally requires knowledge of the constraints, in this case the amount of gas actually available for injection. There are a number of ways to determine gas availability, several of which have been tried as part of the development of the present work.

The first method tried, and perhaps the most obvious one, was to measure the discharge rate coming out of the compressors, which can then be compared with the injection rates going into the wells (Borden et al, 2016). Two major issues arise with this method: tracking and calibration. First, every source and every sink of gas must be accounted for; this includes compressors and gas lift wells, of course, but also any sales lines, interconnections with other compressor stations, other systems fed by the gas lift line, and other lift methods that use gas, such as gas- assisted plunger-lift (GAPL) wells. This complexity limits the rate at which scale can be achieved (a key metric for high-wellcount assets), and also poses data unification issues. Even after every source and sink for gas is identified, and all the data sources are aligned, though, there is an ongoing difficulty associated with keeping every meter calibrated.

The second method tried was storing the capacity of each compressor. This allowed the optimizer to calculate online capacity by knowing which compressors were currently active. The trouble with this method is keeping the capacity data up to date. Of course, each compressor has a nameplate value that it is rated to provide, but those can degrade as time goes on and changes are made to the compressor. The capacities can be updated via a simulation software (similar to the manner in which manufacturers calculate the nameplate ratings), but this process relies on being familiar with exactly what has been done to each compressor, which is not always clearly documented, especially during unplanned outages. (See the discussion section for more commentary on the trade offs between physics-based simulation and data-driven approaches.) Furthermore, this method still relies on a substantial number of measurements, and is thus prone to many of the same issues as the first method.

With these methods attempted and rejected, a new method was created which found more success. Rather than trying to force the rates going to every well to match the capacity, this method treats the system holistically, similar to the way in which a proportional-integral-derivative (PID) controller would. To achieve this, it uses the compressor manifold discharge pressure as a source of feedback information regarding the relative balance between the compression (and any other sources) and the wells (along with any other sinks). If the pressure is low, then the compressors are not keeping up with the injection rate, so injection should be pulled down. On the other hand, if the pressure is suitably high (i.e., reaching the pressure setpoint of the compression), the compressors are keeping up, so injection could be increased if desired. This removes the need for tracking and accurate measurement of every gas flow rate across the field, and leaves in its place a simple requirement of the system pressure and the current setpoints at the wells.

To convert the pressure feedback into a useful constraint, we use a simplifying assumption that over the very short time between optimizations (as little as 15 minutes), the maximum power available to the compressor(s) is constant, as are the factors impacting compression capacity (ambient temperature, physical derating, etc.) with the exception of pressure differential and throughput. The one case where the assumption of stability is clearly invalid is when a compressor shuts down or is restarted during the interval. These cases, however, are simply picked up the next time the optimizer iterates. Next, we assume that the suction pressure coming into the compressor is negligible, and thus that the outlet pressure is roughly equivalent to the pressure differential across the compressor. This allows the following derivation:

$$\begin{gathered} HHP_{t1} = HHP_{t2} \\ HHP_{ti} = \Delta P_{ti} * Q_{ti} \approx P_{out,ti} * Q_{ti} \\ P_{out,t1} * Q_{t1} = P_{out,t2} * Q_{t2} \\ Q_{t2} = \frac{P_{out,t1}}{P_{out,t2}} * Q_{t1} \end{gathered} \tag{1}$$

where HHP_{ti} is the hydraulic horsepower available to (and thus, in the constrained case, utilized by) a compressor, ΔP_{tt} is the pressure differential across the compressor, Q_{ti} is the gas flow rate through the compressor, and $P_{out,ti}$ is the pressure of the gas leaving, each at a specific time *ti*. In the final line, $P_{out,t2}$ is the target pressure, which is generally set slightly below the compressor(s)'s discharge pressure setpoint to make it clear to the optimizer when the compressor has additional capacity and when it is fully employed. This concept is considered further in the throughput maximization subsection in the discussion section.

We extend the derivation with our final assumption, which is that the sum of the injection setpoints is roughly similar to the gas flow rate actually leaving the compressor(s):

$$\begin{gathered} Q_{ti} \approx \sum q_{SP,ti} \\ \sum q_{SP,t2} = \frac{P_{out,t1}}{P_{out,t2}} * \sum q_{SP,t1} \end{gathered} \tag{2}$$

Where $\sum qsp,ti$ is the sum of the gas lift injection rate setpoints of all the wells fed by a given system. This is calculated from measured data the present time (*t1*), while the sum of the setpoints in the future (*t2*) provides the gas availability constraint for the optimizer. This last assumption is the most tenuous of the assumptions by far, but also the least important. All that needs hold true is that the PID-style logic is valid, i.e., that the setpoints have the correct directional impact on the total injection rate. So long as that holds, the system will approach the correct answer, and in most cases does so within a few iterations.

Gas Allocation & Prioritization

Once the total gas available (i.e., the gas availability constraint) has been estimated, gas must be allocated accordingly. As with determining the gas available, there are a number of ways to allocate the gas, each with their own drawbacks.

There is, of course, the trivial solution, in which no setpoints are changed; of course, if the setpoints sum up to a value greater than the gas being compressed, at least one setpoint will not be reached. Typically, the wells that lose injection first are those with the highest casing pressures, which tend to be the stronger producers. Clearly, this is not an optimal solution; we would rather weaker producers miss out on gas first.

To avoid this result, some operators use up a simple shed list (Masud et al, 2023), often implemented manually but sometimes automatically, which typically cuts gas injection to the weakest producers first. This is generally a better solution than the trivial one, but it holds its own issue, because oil production is not directly proportional to gas injection across a series of wells. Because of this, this type of shed list might cause two wells that each use 300 kscf of gas to produce 100 barrels of oil each day would be cut before a well that uses 700 kscf to produce 110 barrels of oil each day. In this situation, 200 barrels of oil would be lost to reduce injection by 600 kscf/d, despite the fact that the third well is clearly using its gas less efficiently.

One could, therefore, set up a system that ranks wells based on the ratio of their oil production to gas injection. This concept, however, ignores one final issue: For an individual well, the marginal return for each additional unit of gas is not constant; rather, it diminishes as the well is given more and more gas, as shown in Fig. 1.

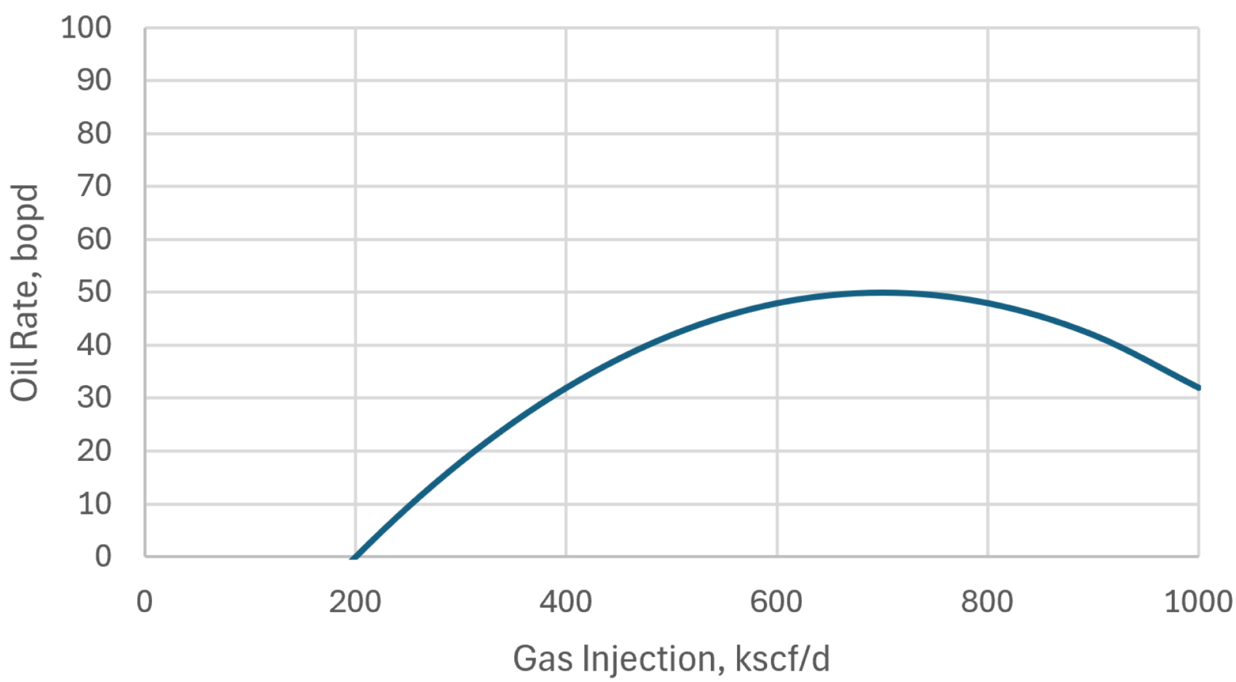

Figure 1—A typical gas lift performance curve (GLPC) describes the relationship between the gas injected into a well and the oil rate that a well will produce at that injection rate.

Fortunately, the gas lift performance curve (GLPC) provides the information to generate the optimal solution. Specifically, the optimal solution in a constrained case is the one where the additional marginal oil for an additional unit of gas (i.e., the slope of the GLPC) is equivalent across all wells and the gas is fully allocated, or no additional benefit can be derived from more injection at any well. There is one caveat here; it is assumed that all wells must receive at least the minimum stable rate (MSR), which is the minimum required to maintain non-zero production, and can be visually identified as the left-most x-intercept of the GLPC. Adding this limitation to the optimization algorithm prevents it from allowing wells to stop producing and load up, which could introduce costs due to unloading that would be difficult to quantify without a much more complex (and thus slower) process.

One additional requirement applied to the optimizer is that it must produce setpoints rounded to the nearest 25kscf/d. This was originally implemented to ease troubleshooting on the part of field personnel but was later found to also serve as a convenient dampening system to avoid very small changes associated with noise in pressure data in the constraint definition step. Where the MSR assumption simplified the algorithm substantially, the rounding requirement complexified it somewhat, changing it from a straightforward quadratic optimization to a discrete optimization.

The process by which this is achieved is shown graphically in Fig. 2. First, the gas lift performance curve for each well in the compressor/well system is obtained via the methods described by Movahed et al in 2024. Next, the GLPCs are broken into chunks (of 100 kscf/d in the figure, for legibility), and the additional marginal oil (i.e., slope) across each step is tabulated. The tables are then combined into a single list, and prioritized from most to least additional oil, with the MSRs for each well at the top. A cumulative total injection rate is then calculated for each row.

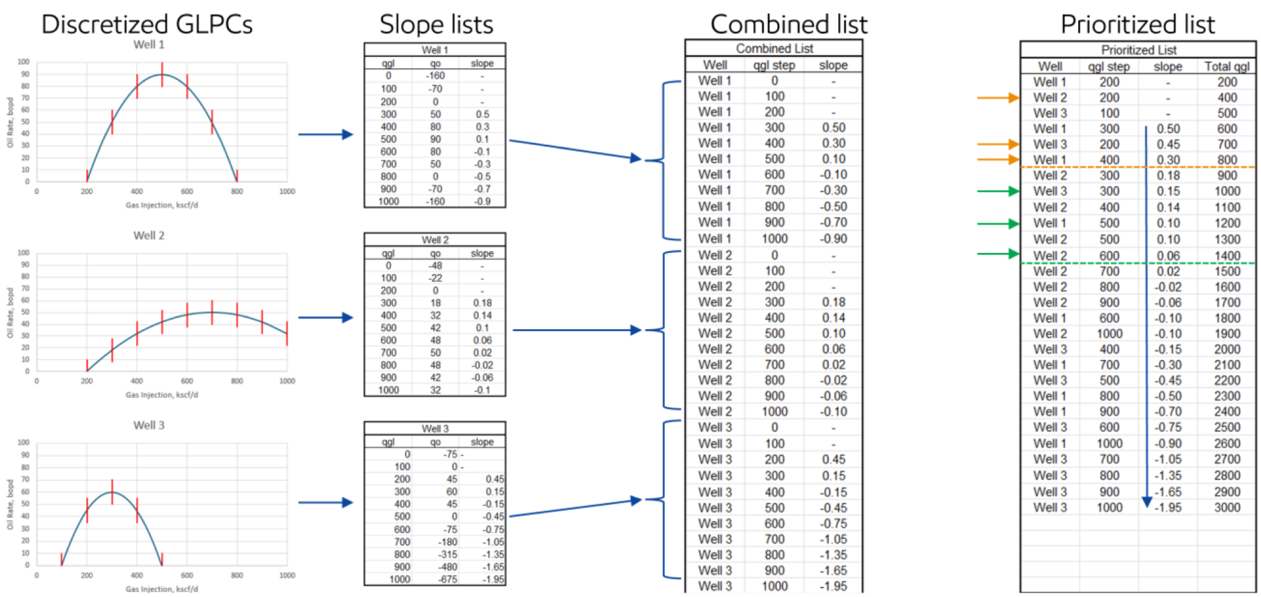

Figure 2—A prioritized list describing how to allocate gas to all wells for any amount of gas available for injection is constructed by discretizing the GLPCs for each well into a set of discrete sections, combining those sets, and ordering the sections by the additional marginal production associated with them.

With the prioritized list created, it becomes very simple to determine the optimal injection rates for any available constraint. The software simply drops the rows where the cumulative total injection is greater than the constraint or where the slope is negative (indicating injection beyond the optimal rate), then takes the largest step reached for each well. For example, in Fig. 2, if the hypothetical system had 850 kscf/d available, the optimal rates (again, enforcing the 100 kscf rounding rule for the figure) would be 400, 200, and 200 for wells 1, 2, and 3 respectively. If the constraint increased to 1425, as might happen after a compressor was restarted, the optimal rates would jump to 500, 600, and 300. In some cases, there may be substantial additional gas (if 2000 kscf/d were available in the example). This is not an issue; the software simply allocates the unconstrained optimal value to each well, avoiding wasted compression costs while simultaneously maximizing production.

Performing the analysis in this manner is beneficial because it allows for very fast responses, even for much larger well counts, which is crucial when optimizing transient processes. The prioritized list is also very similar to a more typical shed list, which field personnel are generally familiar with. This helps improve trust in the software, and thus uptake.

Final Checks and Implementation

The software continuously monitors the pressure data at dozens of compression systems, watching for deviations that are large enough to drive a setpoint change. When it identifies a candidate system in need of optimization, it checks that no optimization controls have been sent out recently (i.e., in the past 15 minutes) to avoid duplicate controls. If no recent changes have been made, it makes the changes at the wells.

Results

Uplift

The calculation of uplift for most optimizations is simple; one need simply find the difference between what happened and what would have happened had no optimization been made. Typically, this involves projecting the preoptimization production forward past the point in time where the optimization was done, and comparing that projection to what actually occurred. Unfortunately, this is a rather difficult proposal during a transient period, which by its very nature invalidates extrapolation from a preceding steady period, both because the system has changed (typically due to a change in gas availability), and because the production expected from the alternative case (if no optimization has been done) is uncertain. Accurately estimating production from the alternate case in this way would likely require a model more complex than the one under investigation, so the idea was rejected. Instead of trying to capture the uplift associated with each individual change, we took a macroscopic approach, in which we considered how well the system was optimized as a whole. Conveniently, this aligns with the goal of optimizing the entire system, and also generates some useful metrics which, to our knowledge, have not been considered previously.

The metric that forms the backbone of the uplift calculation is the ratio of the expected production from the system to the maximum possible production at that time, if everything had been perfectly optimized. We call this the "optimization achievement"; it can be conceptualized as a score for how well the system was optimized at a given point in time. In an ideal world with infinite, perfectly accurate data, we would simply use the measured oil production rates from the wells in the system to calculate the production; however, this level of data is not available, so we use the GLPC for each well to get production expectations from the measured gas injection rates through time, which we have consistent, high-frequency data for. To calculate the maximum possible production, we sum the gas injection rate measurements for the entire system, and then determine what the optimal injection rates for each well would have been given that amount of gas, following the same method described in the Gas Allocation & Prioritization subsection. These rates are then fed back through the GLPC for their respective wells to determine the optimal oil rates, the total of which is the maximum possible production.

The key benefit of using the optimization achievement is that it removes the impact of changes in gas availability on total production. This is exemplified by Fig. 3: the blue line is the gas available at a hypothetical system; the black is the expected production based on the actual injection rates, while the green is the maximum possible production using the same amount of gas with optimal allocation; the vertical yellow line indicates the point in time when the software was rolled out. Considering the left-side plot, it is very hard to say whether any improvement has been made. In the right-side plot, which shows the same period, but with the production data in terms of optimization achievement, it's immediately clear not only that an improvement has been made, and also how substantial that improvement has been.

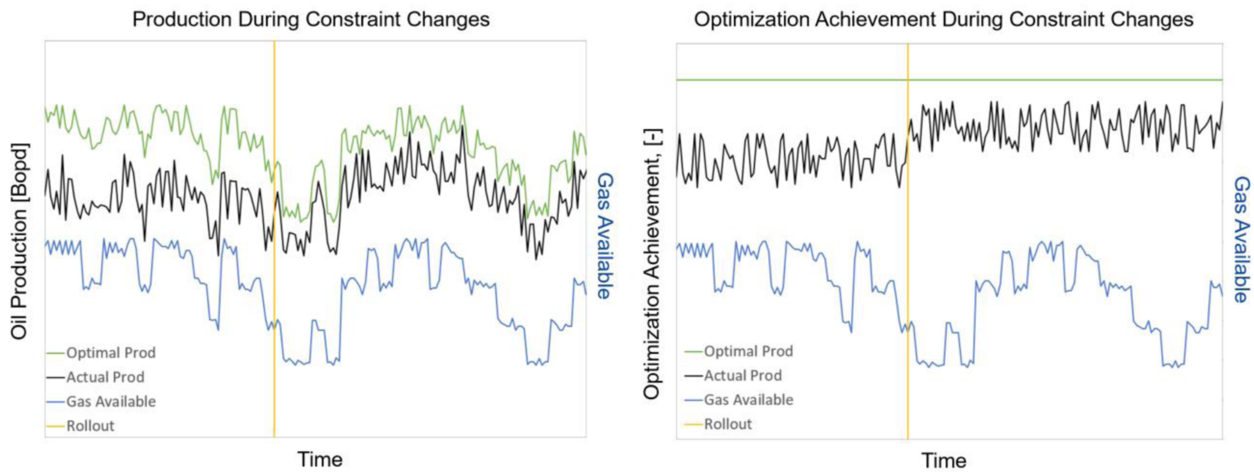

Figure 3—The "Optimization Achievement" quantity provides a metric for how well optimization is being done that isn't impacted by variations in constraints.

To begin the uplift calculation, the optimization achievement is calculated as described above for a given compressor/well system using a range of data overlapping the rollout of the software. For the purpose of this paper, the period of interest was defined as 60 days before the rollout date through 60 days after it. Once the optimization achievement has been calculated, an average optimization value for the portion of time before the rollout is calculated, as indicated by the solid red line in the left-side plot in Fig. 4. This is taken to be a reasonably representative value of the typical optimization achievement for the facility. The pre-rollout average is then projected forward past the rollout date, in the same manner that a simpler uplift calculation would be performed, as indicated by the dashed red line in the left-side plot in Fig. 4. Next, the optimization achievement projection is converted back to a production rate value by multiplying it by the maximum possible production at each time, depicted by the dashed red line in the right-side plot in Fig. 4. Note that the projected production values in the pre-rollout period match the actual values fairly well; this shows that the projection is reasonably accurate, though as an average it is certainly not perfect. Finally, the uplift is calculated as the difference between the actual production rate and the projected production rate, indicated by the green area in the right-side plot in Fig. 4. While none are shown in this example, it is possible for negative uplift values to occur if the optimizer fails to keep the optimization achievement above the pre-rollout average.

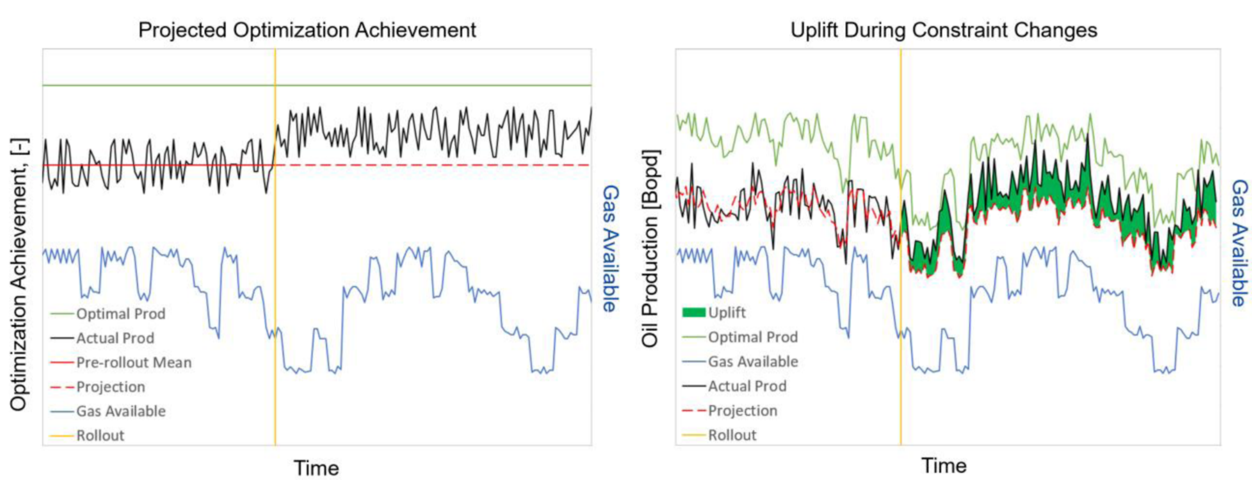

Figure 4—The average pre-rollout Optimization Achievement is used to project expected production, enabling uplift to be calculated as the difference between actual production and the projected value.

As previously mentioned, this software has been applied at dozens of compressor/well systems; the uplift volume calculation described above was performed at each one. A few systems were excluded from the analysis due to a mix of bad/missing data and coincidental changes that invalidated the assumption of consistency underlying the projection, such as large numbers of new wells coming online, or a substantial change to the frequency of compression outages. The uplift volumes from the rest of the compressor/well systems were added together, and then divided by the total projected volume during the same periods, effectively creating a single production-weighted average uplift percentage across all facilities with valid data and no confounding events; the resulting value was an uplift of 3.6% of the total production for the associated wells for the 60 days analyzed. It should be noted that this uplift is in addition to the existing single-well optimization system that was already in place and credited with approximately 2% uplift beyond the manual process previously used to handle optimization.

Pressure Stabilization

Because the optimizer matches injection with compression capacity by watching the discharge pressure, it serves as a natural stabilizer for both the compressor discharge pressure (as shown in Fig. 5) and the injection pressure at each well. This has two main benefits. First, it prevents a process known as cascading failures, compressor shut-downs lead to drops in discharge pressure, which trips additional compressors, causing a cycle that generally leads to a facilitywide shut down. Second, by stabilizing the pressure feeding all wells, not just the gas lift ones, it improves the injection reliability at other wells, such as those using gas-assisted plunger lift (GAPL). Neither of these benefits are considered in the previously discussed uplift calculation, as it focuses purely on the efficiency with which the available gas was utilized at gas lift wells, not on improvements to gas availability or injection at non-gas-lift wells.

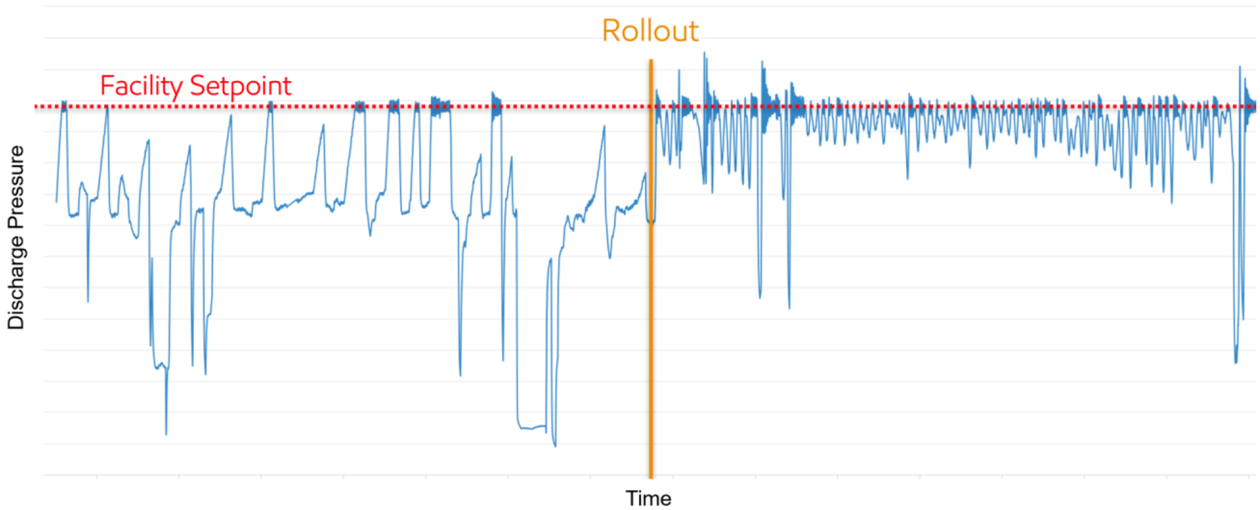

Figure 5—Discharge Pressure Stabilization

Throughput Maximization

Because the software identifies the capacity via the pressure and setpoints it's capable of maintaining, rather than relying on a nameplate or simulation result, it regularly identifies cases where operations personnel were excessively conservative in their estimation of the gas available. This is clearest when the software is first rolled out to a system, and it increases total injection for a system without causing any corresponding drop in compression discharge pressure that would indicate reaching or exceeding a capacity constraint, an example of which is shown in Fig. 6. This maximization of compression capacity utilization is not considered in the uplift calculation described previously, and we have not attempted to calculate a value as it would be a one-time enhancement rather than an ongoing improvement like the main uplift calculation.

That said, the initial uplift is expected to have created significant increases in production in areas where the capacity was significantly underestimated, and thus also under-utilized.

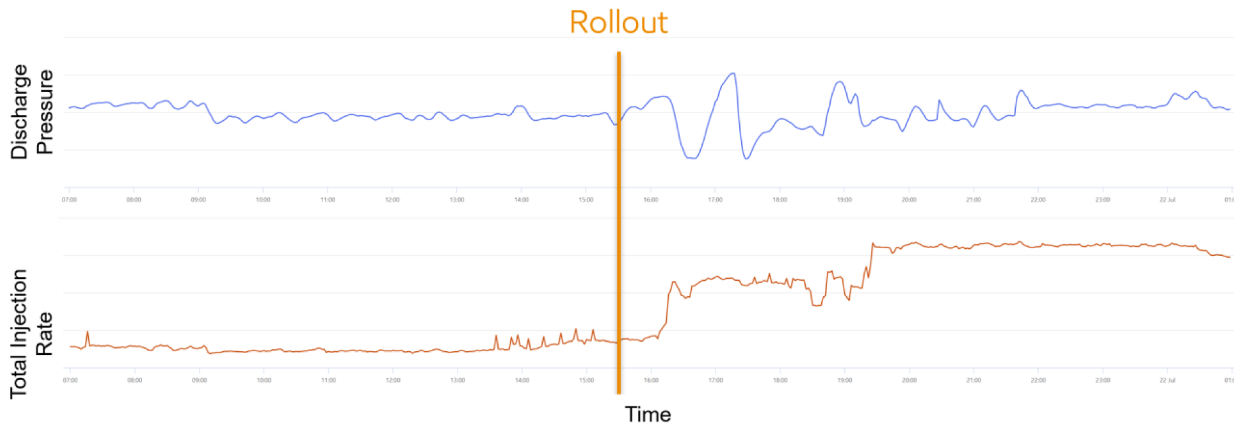

Figure 6—Throughput Increase without Associated Discharge Pressure Drop

Discussion

Physics-based versus Data-driven Processes and Scalability

As outlined in the Methods section, this software relies on data-driven analysis rather than physics-based simulation. Counterintuitively, this significantly reduces the number of required data inputs. This is most evident in the metadata needed to run the software at a given well. While a physics-based model might require detailed inputs such as tubing lengths, diameters, valve parameters, and depths, the data-driven approach described here requires almost none—only the well age is needed, and that is only required if bottom-hole pressure is unavailable.

Moreover, the assumptions required are minimal and relatively relaxed, in contrast to physical simulations, which often depend on numerous assumptions and approximations (e.g., reservoir strength, fluid composition, and selection of appropriate empirical correlations). The data-driven method also eliminates reliance on several well measurements, reducing concerns about data quality and sensor calibration. For this work, and the underlying single-well optimization, the only required time-series inputs are compressor discharge pressure, downhole pressure, well test rates, gas injection rates and setpoints, and choke position (if applicable). In contrast, a physics-based simulation would require additional data such as tubing pressures, compressor discharge rates, and more.

By minimizing data dependencies, the system enables focused calibration of key sensors—far more manageable than maintaining dozens or hundreds of devices. These efficiencies are critical for both initial scalability and longterm operational viability, especially for software intended for autonomous deployment across large assets.

The scalability of this system also benefited from the foundational work done during the development of the single-well optimization workflow, which addressed most data quality and control architecture challenges. As a result, full deployment across dozens of systems and hundreds of wells was completed in just two months, most of which was spent on field engagement and training rather than configuration.

Importance of Autonomy

In applying this software, a common question that was raised was the possibility of applying similar logic in an "open- loop" application, with an operator making the final changes. This was considered, but found to be infeasible due to the frequency and unpredictability of operational constraints. For example, compressor outages can occur multiple times in a single night—an overwhelming scenario for human operators, who must prioritize restoring normal operations over optimization.

This underscores the necessity of autonomous software to manage such dynamic environments. However, operator oversight remains essential. The system includes a manual override mode, originally developed for the singlewell optimizer, which is even more critical here given the optimizer's ability to make frequent adjustments that could override human inputs.

Scope of Control

Despite their capabilities, autonomous systems are limited by the constraints and data they are given. One common concern from field personnel is when the system reduces injection setpoints, based on falling compression, leading to production losses. These concerns typically arose during constrained periods involving compression events.

This highlights the importance of clear communication during rollout—specifically, clarifying what the system can and cannot control. In these cases, we regularly demonstrated that production losses were due to reduced gas availability, not the optimizer's decisions. Building this understanding is key to earning operator trust.

Future work

A current limitation of this system is that it does not account for the impact of choking or changes in flowline backpressure on the gas lift performance curve. A more accurate representation would be a response surface that takes these other factors into account, rather than a simple curve.

We are developing a next-generation system-level optimizer that incorporates such surfaces. This hybrid well model includes three modules: reservoir inflow, wellbore hydraulics, and choke pressure drop. Each module is built and calibrated independently, then integrated to predict oil or liquid rates based on changes in choke setting or gas lift injection rate.

These modules use representative input data or proxies to simplify the workflow. While not always precise, they provide functional forms for pressure and flow relationships that can be calibrated with actual data to yield reliable well-specific predictions. Once validated, these response surfaces will enable a more comprehensive optimizer capable of managing multiple transient constraints simultaneously, as the relationship between production, gas lift injection, choking, and backpressure will be clearly defined.

Conclusions

1. Production is substantially impacted by imperfect gas lift optimization during compression outages, even when single-well optimization systems are actively working, because single-well optimization systems do not capture the system-level constraints associated with compression outages.
2. The production impact of gas lift availability constraints can be mitigated through gas lift changes but requires suitably responsive optimization that considers the broader system of wells.
3. Autonomous control enables responsive optimization, where manual review would inhibit timely responses.
4. Artificial intelligence and machine learning enable system-level optimization in cases where data from some portions of the system is incomplete.
5. Autonomous, multi-well optimization has been shown to produce broad-based uplift of 3.6% of total gas lift production, along with other associated benefits.
6. There remains substantial scope for improvement regarding other constraints and operational variables.

Acknowledgment
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Abbreviations

ESP	- Electrical Submersible Pump
GAPL	- Gas Assisted Plunger Lift
GLPC	- Gas Lift Performance Curve
LightGBM	- Light Gradient-Boosting Machine
PID	- Proportional-Integral-Derivative

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